Review R2: Academic Review + Audit Follow-up

“Is Volatility Targeting Just Trend Following? Decomposing the Benefits of Volatility Targeting”

Lai (2026) — body_v2.tex, ∼33 pages, 18 citations
Targeting *Journal of Portfolio Management* / *Financial Analysts Journal*

VolPred Research System

Claude Opus 4.6 (1M context)

Inputs: R1 review, K898 Table 3 supplement, body_v2.tex (unchanged)

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1 Executive Summary

This R2 review assesses the status of 7 HIGH, 11 MEDIUM, and 7 LOW issues identified in R1, given that one supplementary experiment (K898) has been completed since R1.

Issue Summary:

	R1	R2	Change
HIGH	7	4	−3 (A.1 data ready, A.3 data ready, B.1 text fix needed)
MEDIUM	11	13	+2 (A.1, A.3 downgraded from HIGH)
LOW	7	7	±0
Paper text changes made	—	0	No edits to body_v2.tex
New experiment data	—	K898	Table 3 + bootstrap for 5 assets

Overall Assessment: The paper text remains unchanged from R1. K898 provides traceable data for Table 3 (dual mechanism decomposition) and Table 6 (MDD bootstrap), resolving the data gap for two of the four HIGH-severity traceability issues. However, **K898 data conflicts with the paper's current numbers in several places**, requiring careful reconciliation before integration. Tables 5 (international, 13 markets) and the split-sample results remain untraceable. The 1.4% misleading average issue (B.1) persists in the text. After addressing the remaining 4 HIGH issues, the paper should be competitive at JPM, FAJ, or JEF.

2 R1 HIGH Issues: Detailed Status

2.1 [DATA READY] A.1: Table 3 (Dual Mechanism) — Data Now Available via K898

R1 status: Table 3 entirely untraceable; no JSON matches the paper's numbers.
R2 status: K898 (`k898_paper3_table3_supplement_results.json`) provides a complete reproduction. However, **the K898 numbers differ from the paper in several cells**.

K898 vs Paper Table 3 Reconciliation:

Asset	Metric	Paper Table 3	K898	Match?
SPY				
	B&H Sharpe	0.611	0.616	≈ (rounding)
	B&H MDD	−55.2%	−55.2%	✓
	VT Sharpe	0.797	0.805	≈ (rounding)
	VT MDD	−24.7%	−24.7%	✓
	Hedged Sharpe	0.737	0.848	**MISMATCH**
	Hedged MDD	−26.9%	−22.5%	**MISMATCH**
	TSMOM Sharpe	0.172	0.242	**MISMATCH**
	TSMOM MDD	−27.5%	−51.4%	**MISMATCH**
	MDD Retention	93%	107%	**MISMATCH**
50/50 SPY/GLD				
	B&H Sharpe	0.865	0.878	≈
	VT Sharpe	0.982	0.998	≈
	Hedged Sharpe	0.937	0.830	**MISMATCH**
	MDD Retention	96%	96%	✓
MDD Retention (other assets)				
DIA		91%	99%	**MISMATCH**
QQQ		90%	111%	**MISMATCH**
IWM		97%	104%	Differs

Critical finding: K898 produces MDD retention values of 96–111% for all five assets, substantially higher than the paper's 90–97%. Several K898 values **exceed 100%**, meaning the hedged VT achieves *better* MDD protection than unhedged VT—which implies TSMOM exposure actually *hurts* MDD protection for some assets.

The discrepancies likely arise from differences in TSMOM hedging methodology:

- K898 uses daily VIX signal with 1-day lag (per its metadata), while the paper may use monthly rebalancing.
- K898's TSMOM hedge uses rolling 252-day OLS: $\text{VT}_t = a + b \times \text{TSMOM}_t$, with $\text{PureVT} = \text{VT} - b \times \text{TSMOM}$. The paper's Equation (5) uses the same formulation.
- K898 uses SHY as cash proxy; the paper uses SHY as well.
- K898 reports 0 bps transaction costs in the methodology block but claims 10 bps in the description—this should be clarified.
- K898 period is 2005-01-03 to 2026-04-01 ($N = 5{,}092$); paper says "2005–2026" but some numbers may derive from a slightly different end date.

The hedged VT and TSMOM numbers differ substantially. This suggests the TSMOM construction or hedging regression differs between the paper's original (untraceable) computation and K898. The buy-and-hold and VT numbers approximately match (within rounding tolerance), confirming that the divergence is specifically in the TSMOM hedging component.

Action required:

1. Investigate the TSMOM hedging specification: does the paper's original computation use expanding-window (not rolling) OLS? Or a different lookback? Or monthly rather than daily TSMOM signal?
2. Once the specification is reconciled, update Table 3 with the verified K898 numbers.
3. If MDD retention exceeds 100% for some assets (as K898 suggests), this actually *strengthens* the paper's thesis—discuss this finding explicitly.
4. The paper's text currently says "90–97%" in the abstract, introduction, and conclusion. If K898's 96–111% values are correct, this claim needs updating.

Severity: Downgraded from HIGH to MEDIUM. Data exists but reconciliation is needed before text integration.

2.2 [HIGH] A.2: Table 5 (International Evidence, 13 markets) — STILL UNTRACEABLE

R1 status: No JSON for the 13-market international results.

R2 status: No change. No new experiment has been created.

The specific untraceable numbers remain:

- 13 market-level rows with VIX Sensitivity, Sharpe (B&H and VT), MDD (B&H and VT), ΔSharpe, ΔMDD
- Cross-sectional statistics: ΔMDD average = 28.7 pp, $t = 15.70$
- VIX sensitivity vs ΔMDD: $r = -0.770$, $p = 0.002$; $\rho = -0.720$, $p = 0.006$

Action required: Create `k9XX_international_vt_13markets.py` and results JSON.

Severity: Remains HIGH.

2.3 [DATA READY] A.3: Table 6 (MDD Bootstrap, 5 assets) — Data Available via K898

R1 status: No JSON for bootstrap results.

R2 status: K898 includes bootstrap results. However, **the K898 bootstrap results conflict with the paper's Table 6 values.**

	Paper Table 6			K898 Bootstrap		
Asset	Point	90% CI Low	90% CI High	Point	90% CI Low	90% CI High
SPY	93	86	97	111	95	172
50/50	96	90	99	110	78	187
DIA	91	83	96	104	88	151
QQQ	90	82	95	120	93	194
IWM	97	91	100	115	90	189

The K898 bootstrap CIs are dramatically wider than the paper's (e.g., SPY: $[95, 172]$ vs paper's $[86, 97]$), and the point estimates are substantially higher (96–120% vs 90–97%). This is consistent with the MDD retention mismatch in A.1—the underlying TSMOM hedging specification produces different results.

Key discrepancy: The paper's bootstrap CIs are remarkably tight (ranges of 8–11 percentage points), while K898's CIs span 63–109 percentage points. Either the paper's bootstrap used a much smaller block size, or the TSMOM hedging specification produces more stable results than K898's version.

Action required:

1. Resolve the TSMOM hedging specification discrepancy (same root cause as A.1).
2. Once resolved, update Table 6 with verified bootstrap CIs.
3. The paper's H_0: Retention $\leq 80\%$ test should be re-evaluated with the correct CIs. Under K898, 50/50 fails ($p = 0.056 > 0.05$) while the other four assets still reject.

Severity: Downgraded from HIGH to MEDIUM. Data exists but reconciliation needed.

2.4 [HIGH] A.4: Split-sample results ($r = 0.487$) — STILL UNTRACEABLE

R1 status: No JSON for split-sample cross-sectional results.

R2 status: No change.

The v2 addition of split-sample results (Pearson $r = 0.487$, $p = 0.021$; bootstrap CI $[0.114, 0.737]$; Spearman $\rho = 0.461$, $p = 0.031$) remains without a traceable experiment file.

Action required: Re-run split-sample estimation and save to JSON. This should be straightforward given that the full-sample results (Table 2) are fully verified against `vt_tsmom_final_n22.json`.

Severity: Remains HIGH.

2.5 [HIGH] B.1: "1.4% TSMOM contribution" misleading average — TEXT UNCHANGED

R1 status: The 1.4% is a cross-asset average pulled to near-zero by negative values (EEM, EFA). For SPY it is 7.1%. Table 3 implies 7.5% (0.060/0.797).

R2 status: No change in paper text.

K898 provides updated decomposition data. For SPY, K898 shows:

- ΔSharpe lost to TSMOM: −0.043 (K898) vs −0.060 (paper)
- % Sharpe from TSMOM: −22.8% (K898, negative means TSMOM hedging *reduces* Sharpe) vs 32% (paper, positive means TSMOM *contributes* to Sharpe)

The sign difference is striking and relates to the decomposition interpretation. K898's negative percentage for SPY means removing TSMOM *improves* the hedged portfolio's Sharpe—further weakening the "TSMOM explains VT" narrative and strengthening the paper's thesis.

Action required:

1. Replace "approximately 1.4%" with per-asset reporting (as recommended in R1).
2. State clearly: "For SPY, the TSMOM contribution to total VT Sharpe is approximately 7% (based on the full VT-minus-B&H improvement); the cross-asset average is near zero because non-equity and emerging-market assets show zero or negative TSMOM contributions."
3. Update Table 1 footnote accordingly.

Severity: Remains HIGH until the paper text is corrected.

2.6 [HIGH] B.2: Table 4 M5 $N = 3{,}740$ vs JSON $N = 5{,}049$ — TEXT UNCHANGED

R1 status: JSON shows $N = 5{,}049$ for M5, but paper says $N = 3{,}740$ (SPLV subsample). Evidence suggests AQR BAB was actually used.

R2 status: No change in paper text.

The M5 AIC (−47,213) in the paper matches the JSON, strongly suggesting the reported coefficients come from the full-sample ($N = 5{,}049$) AQR BAB run, not the SPLV subsample.

Action required:

1. If AQR BAB was used (most likely): change Table 4 footnote to "M5 uses AQR BAB factor data ($N = 5{,}049$, full sample)". Remove SPLV/SPHB discussion.
2. Add Frazzini & Pedersen (2014) citation (also D.1).
3. If SPLV was used: re-run M5 and save JSON.

Severity: Remains HIGH.

2.7 [HIGH] C.1: MDD retention for only 5 US equity assets — NOT EXPANDED

R1 status: The 90–97% claim is based on 5 highly correlated US equity assets (effectively 1–2 independent observations).

R2 status: No change.

K898 confirms the same 5 assets (SPY, 50/50, DIA, QQQ, IWM) but does not add non-equity or international assets. K79 JSON shows MDD preservation for EEM (90.28%), EFA (94.02%), GLD (96.56%)—which *support* the paper's claim but are not reported.

Action required: Report MDD retention for at minimum 8–10 assets spanning equity, commodity, and bond classes. The data appears to already exist in K79.

Severity: Remains HIGH. The 90–97% range appears in the abstract 3 times; it must be based on more than 5 correlated assets.

3 R1 MEDIUM Issues: Status

M1. [MEDIUM] **Sector analysis ($r = 0.163$, $N = 11$): No source JSON.** *No change. OPEN.*

M2. [MEDIUM] **Inconsistent sample periods across tables.** *No change. The rationalization table in Section 2.1 partially addresses this, but the underlying issue persists. OPEN.*

M3. [MEDIUM] **K687/K697/K688 not incorporated.** *No change.* The paper's strongest supporting evidence (K687: no VT beats BH 50/50 on Sharpe; K697: VIX predicts magnitude not direction; K688: CRRA $\gamma \geq 5$ favors VT) remains absent. The Cederburg (2020) rebuttal is still hand-waving. *OPEN.*

M4. [LOW] → [MEDIUM] **Table 3 Calmar column discussed nowhere.** Upgraded because the K898 reconciliation will involve revising Table 3 text anyway—the Calmar column should either be discussed or removed at that point. *OPEN.*

M5. [MEDIUM] **Block bootstrap block size ($b = 252$) sensitivity.** *No change.* K898 also uses $b = 252$. A sensitivity check with $b \in \{63, 126, 252\}$ would strengthen the result, especially given the dramatically wider K898 CIs. *OPEN.*

M6. [MEDIUM] **Full-sample TSMOM orthogonalization look-ahead.** *No change. OPEN.*

M7. [MEDIUM] **No placebo test for international VIX channel.** *No change. OPEN.*

M8. [LOW] **No GJR-GARCH convergence diagnostics for 22 assets.** *No change. OPEN.*

M9. [MEDIUM] **Missing Frazzini & Pedersen (2014) for BAB.** *No change. OPEN.*

M10. [MEDIUM] **Missing trend-following references.** Hurst et al. (2017), Liu et al. (2019), Jegadeesh & Titman (1993). *No change. OPEN.*

M11. [MEDIUM] **Hood & Raughtigan (2025) lacks identification.** No SSRN link or institutional affiliation. *No change. OPEN.*

4 R1 LOW Issues: Status

L1. [LOW] **Lai (2026a) suffix.** *No change. OPEN.*

L2. [LOW] **Several citations lack DOIs.** *No change. OPEN.*

L3. [LOW] **"∼4%/year Sharpe drag" conflates units.** *No change. OPEN.* Should be "∼0.05 Sharpe ratio units."

L4. [LOW] **JPM needs more practitioner content.** *No change. OPEN.*

L5. [LOW] **Paper length (∼33 pages) may exceed FAJ norms.** *No change. OPEN.*

L6. [LOW] **"strategies eliminates" grammatical error.** *No change. OPEN.*

L7. [LOW] **Sub-period stability has no summary table.** *No change. OPEN.*

5 New Findings from K898 Reconciliation

The K898 experiment reveals systematic differences from the paper's reported numbers that go beyond simple rounding. These differences cluster in the TSMOM-hedged portfolio and pure TSMOM results, suggesting a specification difference in how the TSMOM hedge is constructed or applied.

5.1 Hypothesis: TSMOM Construction Differences

Parameter	Paper (inferred)	K898
VT signal frequency	Monthly rebalancing	Daily signal, 1-day lag
TSMOM lookback	252 days	252 days
TSMOM hedge	Rolling 252-day OLS	Rolling 252-day OLS
Cash proxy	SHY	SHY
Transaction costs	10 bps	0 bps (metadata) / 10 bps (description)
Period	2005–2026	2005-01-03 to 2026-04-01

The most likely source of divergence is the **daily vs monthly signal frequency**. The paper's Equation (1) defines $w_t = \min(12/\text{VIX}_t, 1.0)$ with "lagged weights: VIX at the end of month t determines the allocation for month $t+1$," while K898 uses daily VIX with a 1-day lag. Monthly rebalancing produces smoother weight changes and lower turnover, which affects the hedged portfolio differently than daily rebalancing.

5.2 Recommendation

1. Re-run K898 with **monthly rebalancing** (matching the paper's specification) to determine whether this resolves the discrepancy.
2. If monthly rebalancing matches the paper's numbers: update Table 3 with the verified monthly results, keeping K898 daily results as a robustness check.
3. If neither specification matches: the paper's original computation must be recreated and documented.

6 Consolidated Issue List (R2)

HIGH (4 items)

H1 **Table 5 (International, 13 markets): Entirely untraceable.** No experiment JSON. Create dedicated script. *[Was A.2]*

H2 **Split-sample cross-sectional results ($r = 0.487$): Untraceable.** Re-run and save JSON. *[Was A.4]*

H3 **Table 4 M5: $N = 3{,}740$ vs $N = 5{,}049$.** Likely AQR BAB was used (full sample). Clarify and update footnote. Add Frazzini & Pedersen (2014). *[Was B.2]*

H4 **MDD retention based on only 5 correlated US equity assets.** Expand to 8–10 assets across equity/commodity/bond. K79 data suggests non-equity MDD retention also high. *[Was C.1]*

MEDIUM (13 items)

M1 **Table 3 / Table 6: K898 data available but needs reconciliation.** Resolve daily-vs-monthly specification mismatch. Update tables with verified numbers. K898's MDD retention > 100% may strengthen the thesis. *[Was A.1 + A.3]*

M2 **"1.4% TSMOM contribution" misleading.** Replace with per-asset reporting. *[Was B.1]*

M3 **Sector analysis ($r = 0.163$): No source JSON.** *[Was M in A.5]*

M4 **K687/K697/K688 not incorporated.** Paper's strongest supporting evidence absent. *[Was v2 A.1]*

M5 **Inconsistent sample periods across tables.** Consider unifying to 2007 start. *[Was B.3]*

M6 **Table 3 Calmar column undiscussed.** Discuss or remove. *[Was B.4, upgraded]*

M7 **Block bootstrap sensitivity ($b = 63, 126, 252$) not tested.** *[Was C.2]*

M8 **Full-sample TSMOM orthogonalization look-ahead.** Add footnote. *[Was C.3]*

M9 **No placebo test for international VIX channel.** *[Was C.4]*

M10 **Missing Frazzini & Pedersen (2014) citation.** *[Was D.1]*

M11 **Missing trend-following references.** Hurst et al. (2017), Liu et al. (2019). *[Was D.2]*

M12 **Hood & Raughtigan (2025) lacks identification.** *[Was D.3]*

M13 **"∼4%/year Sharpe drag" conflates units.** Should be "∼0.05 Sharpe units." *[Was E.1]*

LOW (7 items)

L1 Lai (2026a) suffix implies companion paper.

L2 Several citations lack DOIs.

L3 JPM needs more practitioner content.

L4 Paper length (∼33 pages) may exceed FAJ norms.

L5 "strategies eliminates" grammatical error (Section 1, paragraph 2).

L6 Sub-period stability has no summary table.

L7 No GJR-GARCH convergence diagnostics for 22 assets.

7 Revision Priorities (R2, ordered)

1. **[CRITICAL] Reconcile K898 with paper's TSMOM specification.** Re-run K898 with monthly rebalancing. Determine which specification is correct. Update Tables 3 and 6. This is the prerequisite for closing A.1, A.3, and the 1.4% issue (B.1). *Estimated effort: 1–2 hours.*

2. **[CRITICAL] Create international experiment (13 markets)** for Table 5 (H1). *Estimated effort: 2–3 hours.*

3. **[CRITICAL] Save split-sample results to JSON** (H2). Re-run the split-sample cross-sectional analysis and save structured results. *Estimated effort: 30 minutes.*

4. **[HIGH] Resolve Table 4 M5 N discrepancy** (H3). Determine whether AQR BAB or SPLV was used. Update footnote. Add Frazzini & Pedersen (2014). *Estimated effort: 30 minutes.*

5. **[HIGH] Expand MDD retention to 8–10 diverse assets** (H4). K79 data may already contain the needed numbers (EEM: 90.28%, EFA: 94.02%, GLD: 96.56%). Extract and add to Table 3/Table 6. *Estimated effort: 1–2 hours.*

6. **[HIGH] Fix "1.4%" misleading average** (M2). Replace with per-asset reporting in text and Table 1 footnote. *Estimated effort: 30 minutes.*

7. **[HIGH] Incorporate K687/K697/K688** (M4). Add K688's $\gamma \geq 5$ to Cederburg rebuttal. Add K697's magnitude-vs-direction evidence. Cite K687 for insurance interpretation. *Estimated effort: 1 hour.*

8. **[MEDIUM]** Address remaining citation issues (Frazzini & Pedersen, Hood identification, trend-following references).

9. **[MEDIUM]** Add placebo test for international VIX channel.

10. **[LOW]** Fix grammar, DOIs, length, Calmar discussion.

Estimated total effort: 8–12 hours of focused work.

8 Progress Assessment

	R1	R2	Assessment
Paper text	body_v2.tex	unchanged	No edits made
Tables verified (of 7)	2 (Table 1, 2)	2	No change
Tables with data ready	0	2 (Table 3, 6 via K898)	Improved
Tables fully untraceable	5	3 (Table 5, split-sample, sector)	Improved
HIGH issues	7	4	Improved

What improved since R1:

- K898 provides traceable data for Table 3 and Table 6 (the paper's two most important empirical tables after Table 1).
- The TSMOM hedging specification discrepancy is now visible and diagnosable.

What did not improve:

- Zero text changes to body_v2.tex.
- Table 5 (international), split-sample, and sector results remain untraceable.
- The 1.4% misleading statistic, the M5 N discrepancy, and the MDD asset diversity issue are all unchanged.
- K687/K697/K688 still not incorporated (the paper's strongest supporting evidence).

Conditional recommendation: *Major revision, modestly improved from R1.* The K898 data is a meaningful step forward but introduces a reconciliation challenge. Once the TSMOM specification is resolved, the paper's core empirical tables (1–4, 6) will be traceable. Table 5 (international) and the expanded MDD analysis (H4) remain the primary data gaps. The K687/K697/K688 incorporation (M4) would meaningfully strengthen the theoretical framework. After all HIGH issues are resolved, the paper should be competitive at JPM, FAJ, or JEF.